Differentiation

ANSWER KEY

The chain rule

1 Find $\frac{dy}{dx}$ for $y = (3x+1)^5$.
$\frac{dy}{dx} = 5(3x+1)^4 \cdot 3 = 15(3x+1)^4$.

2 Find $\frac{dy}{dx}$ for $y = \sqrt{4x^2+1}$.
Write $y = (4x^2+1)^{1/2}$. $\frac{dy}{dx} = \frac{1}{2}(4x^2+1)^{-1/2} \cdot 8x = \frac{4x}{\sqrt{4x^2+1}}$.

3 Find $\frac{dy}{dx}$ for $y = (2x-5)^6$.
$\frac{dy}{dx} = 6(2x-5)^5 \cdot 2 = 12(2x-5)^5$.

The product rule

4 Find $\frac{dy}{dx}$ for $y = x^2 \sin x$.
$\frac{dy}{dx} = 2x\sin x + x^2\cos x$.

5 Find $\frac{dy}{dx}$ for $y = e^{3x}\cos x$.
$\frac{dy}{dx} = 3e^{3x}\cos x + e^{3x}(-\sin x) = e^{3x}(3\cos x - \sin x)$.

The quotient rule

6 Find $\frac{dy}{dx}$ for $y = \frac{x^2}{2x+1}$.
$\frac{dy}{dx} = \frac{2x(2x+1) - x^2(2)}{(2x+1)^2} = \frac{4x^2+2x-2x^2}{(2x+1)^2} = \frac{2x^2+2x}{(2x+1)^2} = \frac{2x(x+1)}{(2x+1)^2}$.

7 Find $\frac{dy}{dx}$ for $y = \frac{\sin x}{x}$.
$\frac{dy}{dx} = \frac{x\cos x - \sin x}{x^2}$.

8 Find $\frac{dy}{dx}$ for $y = \frac{e^x}{x^2+1}$.
$\frac{dy}{dx} = \frac{e^x(x^2+1) - e^x(2x)}{(x^2+1)^2} = \frac{e^x(x^2-2x+1)}{(x^2+1)^2} = \frac{e^x(x-1)^2}{(x^2+1)^2}$.

Implicit differentiation

9 Find $\frac{dy}{dx}$ given $x^2 + y^2 = 25$.
Differentiate both sides with respect to x: $2x + 2y\frac{dy}{dx} = 0$, so $\frac{dy}{dx} = -\frac{x}{y}$.

10 Find $\frac{dy}{dx}$ given $x^3 + y^3 = 6xy$, and evaluate it at the point $(3, 3)$.
Differentiate: $3x^2 + 3y^2\frac{dy}{dx} = 6y + 6x\frac{dy}{dx}$. Rearranging: $\frac{dy}{dx}(3y^2 - 6x) = 6y - 3x^2$, so $\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}$. At $(3, 3)$: $\frac{2(3) - 9}{9 - 2(3)} = \frac{-3}{3} = -1$.

11 Find $\frac{dy}{dx}$ given $y^2 - xy + x^2 = 7$, and evaluate it at the point $(1, 3)$.
Differentiate: $2y\frac{dy}{dx} - \left(y + x\frac{dy}{dx}\right) + 2x = 0$. Rearranging: $\frac{dy}{dx}(2y - x) = y - 2x$, so $\frac{dy}{dx} = \frac{y-2x}{2y-x}$. At $(1, 3)$: $\frac{3-2}{6-1} = \frac{1}{5}$.

12 Find $\frac{dy}{dx}$ given $\sin(xy) = x$, using implicit differentiation.
Differentiate both sides: $\cos(xy)\left(y + x\frac{dy}{dx}\right) = 1$. Expanding: $y\cos(xy) + x\cos(xy)\frac{dy}{dx} = 1$. Rearranging: $\frac{dy}{dx} = \frac{1 - y\cos(xy)}{x\cos(xy)}$.

Parametric differentiation

13 A curve is defined parametrically by $x = t^2$, $y = 2t^3$. Find $\frac{dy}{dx}$ in terms of t.
$\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 6t^2$. $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{6t^2}{2t} = 3t$ (for $t \neq 0$).

14 A curve is defined by $x = 3\cos t$, $y = 3\sin t$. Find $\frac{dy}{dx}$ in terms of t, and find the gradient at $t = \frac{\pi}{4}$.
$\frac{dx}{dt} = -3\sin t$, $\frac{dy}{dt} = 3\cos t$. $\frac{dy}{dx} = \frac{3\cos t}{-3\sin t} = -\cot t$. At $t = \frac{\pi}{4}$: $-\cot\left(\frac{\pi}{4}\right) = -1$, since $\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right)$.

Related rates

15 A spherical balloon is inflated so its volume increases at a constant rate of 100 cm^3/s. Find the rate of increase of the radius when $r = 5$ cm, using $V = \frac{4}{3}\pi r^3$, to three decimal places.
$\frac{dV}{dr} = 4\pi r^2$. At $r = 5$: $\frac{dV}{dr} = 4\pi(25) = 100\pi$. $\frac{dr}{dt} = \frac{dV/dt}{dV/dr} = \frac{100}{100\pi} = \frac{1}{\pi} \approx 0.318$ cm/s.

16 The area of a circular ripple increases at 8π cm^2/s. Find the rate of change of the radius when $r = 4$ cm, using $A = \pi r^2$.
$\frac{dA}{dr} = 2\pi r$. At $r = 4$: $\frac{dA}{dr} = 8\pi$. $\frac{dr}{dt} = \frac{dA/dt}{dA/dr} = \frac{8\pi}{8\pi} = 1$ cm/s.

17 A cone-shaped tank (vertex down) has height 10 cm and top radius 4 cm, so at water depth h the radius is $r = \frac{2h}{5}$. Water is poured in at 12 cm^3/s. Find the rate of rise of the water level when $h = 5$ cm, using $V = \frac{1}{3}\pi r^2 h$.
Substitute $r = \frac{2h}{5}$: $V = \frac{1}{3}\pi\left(\frac{2h}{5}\right)^2 h = \frac{4\pi}{75}h^3$. $\frac{dV}{dh} = \frac{12\pi}{75}h^2 = \frac{4\pi}{25}h^2$. At $h = 5$: $\frac{dV}{dh} = \frac{4\pi}{25}(25) = 4\pi$. $\frac{dh}{dt} = \frac{dV/dt}{dV/dh} = \frac{12}{4\pi} = \frac{3}{\pi} \approx 0.955$ cm/s.

18 The radius of a circular oil slick increases at a constant rate of 0.5 m/min. Find the rate of increase of its area when $r = 20$ m.
$A = \pi r^2$, so $\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$. At $r = 20$: $\frac{dA}{dt} = 2\pi(20)(0.5) = 20\pi \approx 62.83$ m^2/min.

Differentiating trigonometric, exponential and logarithmic composites

19 Find $\frac{dy}{dx}$ for $y = \sin(3x^2)$.
$\frac{dy}{dx} = \cos(3x^2) \cdot 6x = 6x\cos(3x^2)$.

20 Find $\frac{dy}{dx}$ for $y = \ln(5x^2 + 1)$.
$\frac{dy}{dx} = \frac{10x}{5x^2 + 1}$.

21 Find $\frac{dy}{dx}$ for $y = e^{\sin x}$.
$\frac{dy}{dx} = \cos x \cdot e^{\sin x}$.

22 Find $\frac{dy}{dx}$ for $y = \ln(\sin x)$.
$\frac{dy}{dx} = \frac{\cos x}{\sin x}$.

Applying the rules: tangents and second derivatives

23 Find the equation of the tangent to $y = xe^{-x}$ at $x = 1$.
By the product rule: $\frac{dy}{dx} = e^{-x} + x(-e^{-x}) = e^{-x}(1 - x)$. At $x = 1$: $\frac{dy}{dx} = e^{-1}(0) = 0$. Since $y(1) = e^{-1}$, the tangent is horizontal: $y = \frac{1}{e}$.

24 Find $\frac{d^2y}{dx^2}$ for $y = \ln(x^2 + 1)$, and evaluate it at $x = 1$.
$\frac{dy}{dx} = \frac{2x}{x^2 + 1}$. By the quotient rule: $\frac{d^2y}{dx^2} = \frac{2(x^2 + 1) - 2x(2x)}{(x^2 + 1)^2} = \frac{2 - 2x^2}{(x^2 + 1)^2}$. At $x = 1$: $\frac{2 - 2}{4} = 0$.

A well-designed word problem: a ladder sliding down a wall

25 This question has three parts. A ladder 5 m long leans against a vertical wall, its base on horizontal ground. The base slides away from the wall at a constant rate of 0.4 m/s. (a) Let x be the distance of the base from the wall and y the height of the top of the ladder. Write the relationship between x and y, and differentiate implicitly with respect to time t to relate $\frac{dx}{dt}$ and $\frac{dy}{dt}$. (b) Find the rate at which the top of the ladder is sliding down the wall when the base is 3 m from the wall. (c) Using $x = 5\cos\theta$, where θ is the angle between the ladder and the ground, find the rate at which θ is changing at that instant.
(a) $x^2 + y^2 = 25$. Differentiating with respect to t: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$, so $\frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$. (b) When $x = 3$, $y = \sqrt{25 - 9} = 4$. $\frac{dy}{dt} = -\frac{3}{4}(0.4) = -0.3$ m/s, so the top slides down at 0.3 m/s. (c) Differentiating $x = 5\cos\theta$: $\frac{dx}{dt} = -5\sin\theta\frac{d\theta}{dt}$. At this instant $\sin\theta = \frac{y}{5} = \frac{4}{5} = 0.8$, so $0.4 = -5(0.8)\frac{d\theta}{dt} = -4\frac{d\theta}{dt}$, giving $\frac{d\theta}{dt} = -0.1$ rad/s (the angle is decreasing).